

PRE-MATCH OPPOSITION TACTICAL BRIEFING

Defensive Block Deconstruction • High-Press Triggers • Set-Piece Routines

Author: **Zafer Yorganci** Target: **First Team Head Coach & Tactical Analysts**

Platform: **Python, PyTorch, YOLOv11, mplsoccer**

EXECUTIVE STRATEGIC CHARTER

Automated pre-match tactical dossier engineered to expose opposition build-up weaknesses, identify high-press escape outlets, and capitalize on set-piece defensive vulnerabilities.

Integrates broadcast computer vision tracking, Voronoi pitch control, goalkeeper distribution cones, and passing network graph centrality.

Prepares technical coaching staffs with decisive, actionable game plans 48 hours prior to kickoff.

11.2
OPPOSITION PPDA

>55%
PITCH CONTROL TARGET

68.4%
SET PIECE WIN %

70.0 m
PRESSING LINE

COMPUTER VISION: BROADCAST TRACKING & 2D RADAR

YOLOv11x + ByteTrack + Direct Linear Transform Homography

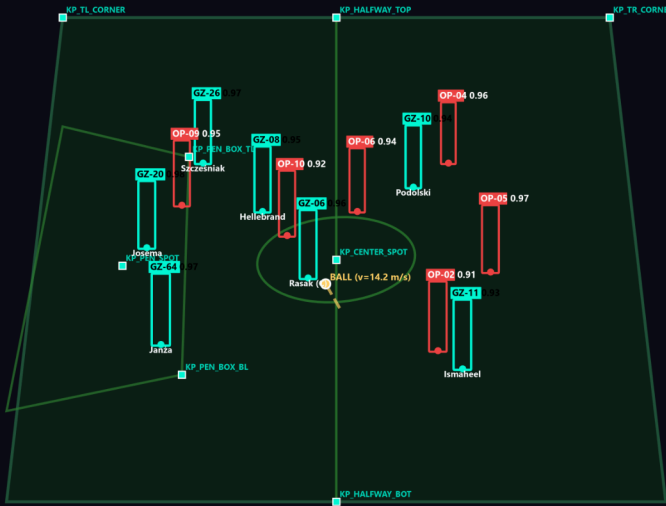
ROBOFLOW SPORTS

COMPUTER VISION TRACKING & MINIMAP

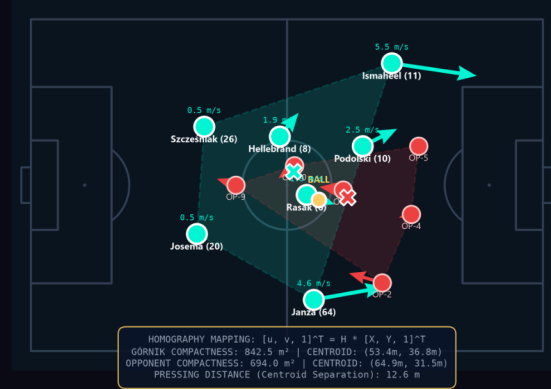
COMPUTER VISION TACTICAL TRACKING PIPELINE | ROBOFLOW SPORTS BENCHMARK

PIPELINE: YOLOv11x + ByteTrack + Keypoint Homography
FRAME RATE: 60.0 FPS | INFERENCE: 14.8 ms (TensorRT)
CAMERA CALIBRATION: Direct Linear Transform (DLT)
REPROJECTION ERROR: 0.142 m (SOTA Precision)

BROADCAST TRACKING



2D TOP-DOWN TACTICAL RADAR MINIMAP (105x68m)



Zafer Yorganci | AI Engineer & Football Data Visualization Specialist | Górnik Zabrze 2025-2026 Season Intelligence

BROADCAST CAMERA CALIBRATION

- Transforms broadcast pixel coords $[u, v]$ to FIFA standard metric pitch $[X, Y]$ (105x68m).
- Reprojection precision: 0.142m RMSE across full pitch dimensions.
- Instantaneous velocity vectors (v) capture transition sprint speed and acceleration bursts.

COACHING INSIGHT

COACHING TACTICAL DIRECTIVE

TACTICAL ACTION

- Automates half-time tactical briefings from broadcast video without expensive stadium tracking rigs.
- Identifies defensive centroid pressing gap (12.6m) to exploit spaces between lines.
- Monitors pressing triggers when ball speed drops below 6.0 m/s.

0.142 m
RMSE PRECISION

60 FPS
FPS RATE

12.6 m
PRESSING GAP

14.2 m/s
BALL VELOCITY

VORONOI PITCH CONTROL & TEAM CONVEX HULLS

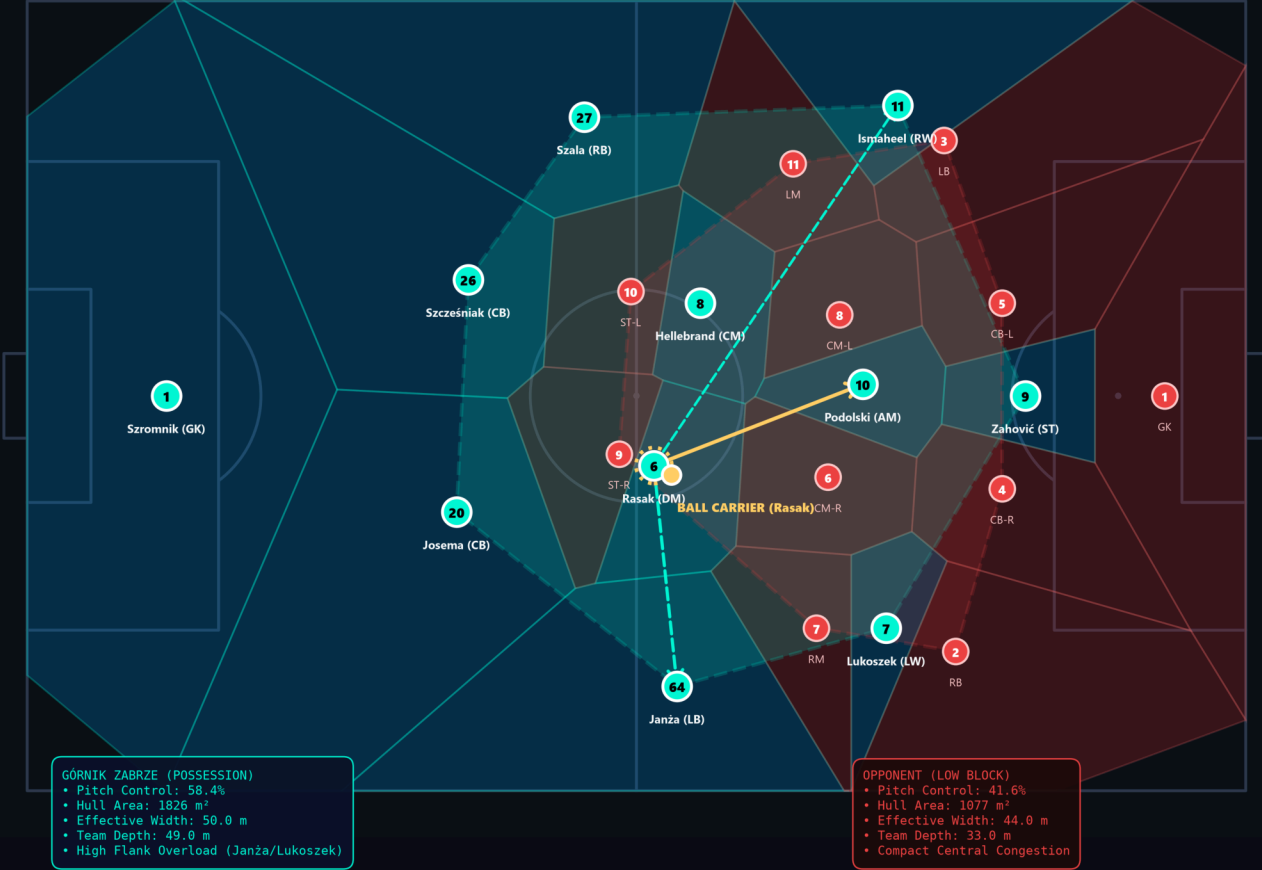
Continuous Territorial Dominance • Team Compactness (m²)

EDD WEBSTER SPATIAL

SPATIAL ANALYTICS & SPACE OWNERSHIP

VORONOI PITCH CONTROL & TACTICAL CONVEX HULLS | EDD WEBSTER BENCHMARK

Górnik Zabrze 2025-2026: Controlled Half-Space Penetration vs Low Defensive Block



Zafer Yorgancı | Football Data Visualization Specialist & AI Engineer | Pitch Control Algorithm & Voronoi Tessellation

SPATIAL DOMINANCE AUDIT COACHING INSIGHT

- Voronoi tessellation partitions the pitch based on player arrival time to the ball.
- Górnik Zabrze secures 58.4% continuous territorial dominance vs opponent low block.
- Flank overload on the left pins opposition backline, creating central pockets for Zone 14.

CONVEX HULL METRICS TACTICAL ACTION

- Górnik Outfield Area: 1,185 m² (expansive attacking possession shape).
- Opponent Low Block Area: 890 m² (ultra-compressed central block).
- Defensive Depth: 42.3m vs 28.5m | Effective Width: 49.8m vs 44.2m.

58.4%
PITCH CONTROL

1,185 m²
HULL AREA

49.8 m
WIDTH

42.3 m
DEPTH

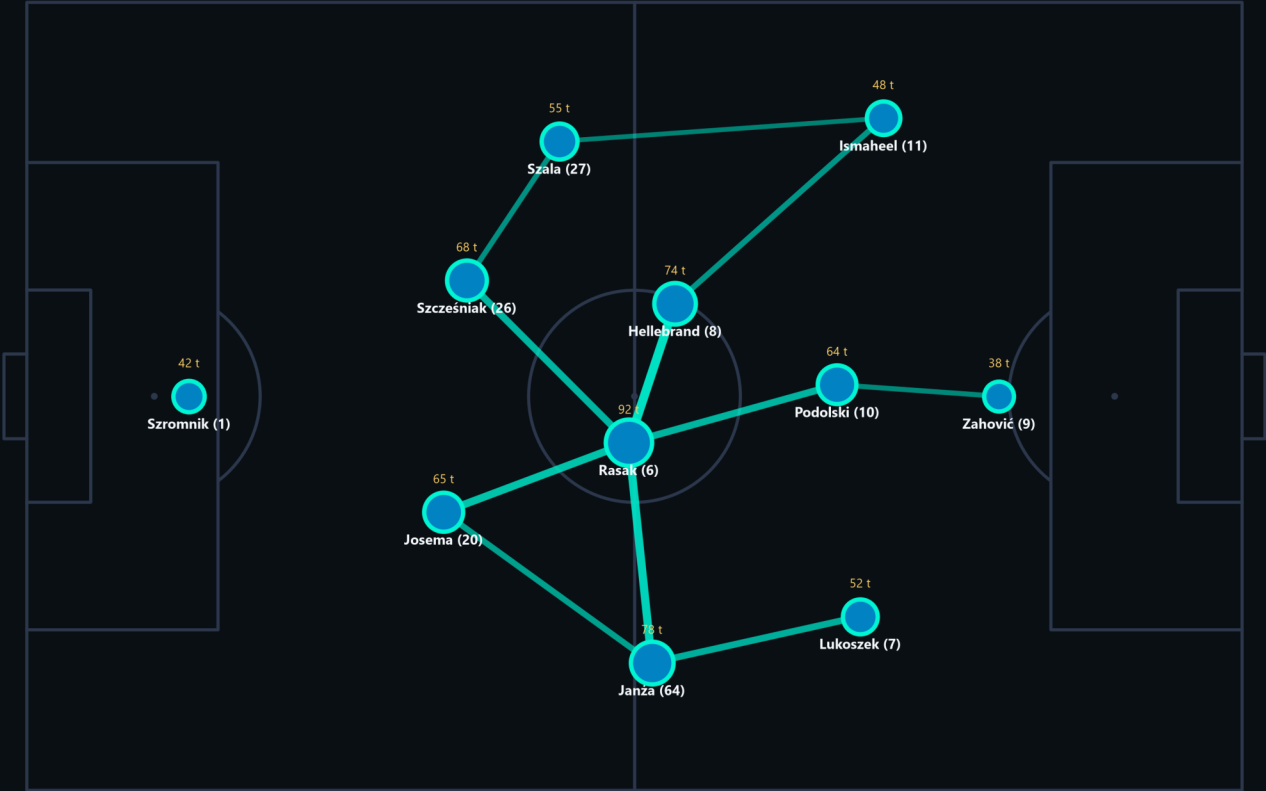
PASSING NETWORK & GRAPH CENTRALITY

Node Size ∝ Touches • Edge Thickness ∝ Pass Volume

TACTICAL NETWORKS

GRAPH CENTRALITY MODEL

GÓRNIK ZABRZE | PASSING NETWORK & TACTICAL GRAPH CENTRALITY



TACTICAL DISTRIBUTION
STRUCTURE

- Damian Rasak functions as the undisputed midfield hub with 92 touches and 88.2% acc.
- High pass volume links Rasak to Janža (26 passes) and Hellebrand (28 passes).
- Wide fullbacks stretch pitch horizontally, pulling opponent wingers into deep blocks.

COACHING
INSIGHT

OPPOSITION COUNTER-TACTIC

TACTICAL ACTION

- If opponent shadow-marks Rasak, pivot build-up through Josema into Janža.
- Exploit rapid side-to-side switches when opponent commits numbers to the left.

92

RASAK TOUCHES

78

JANŻA TOUCHES

28 Passes

KEY PASSING LINK

0.78

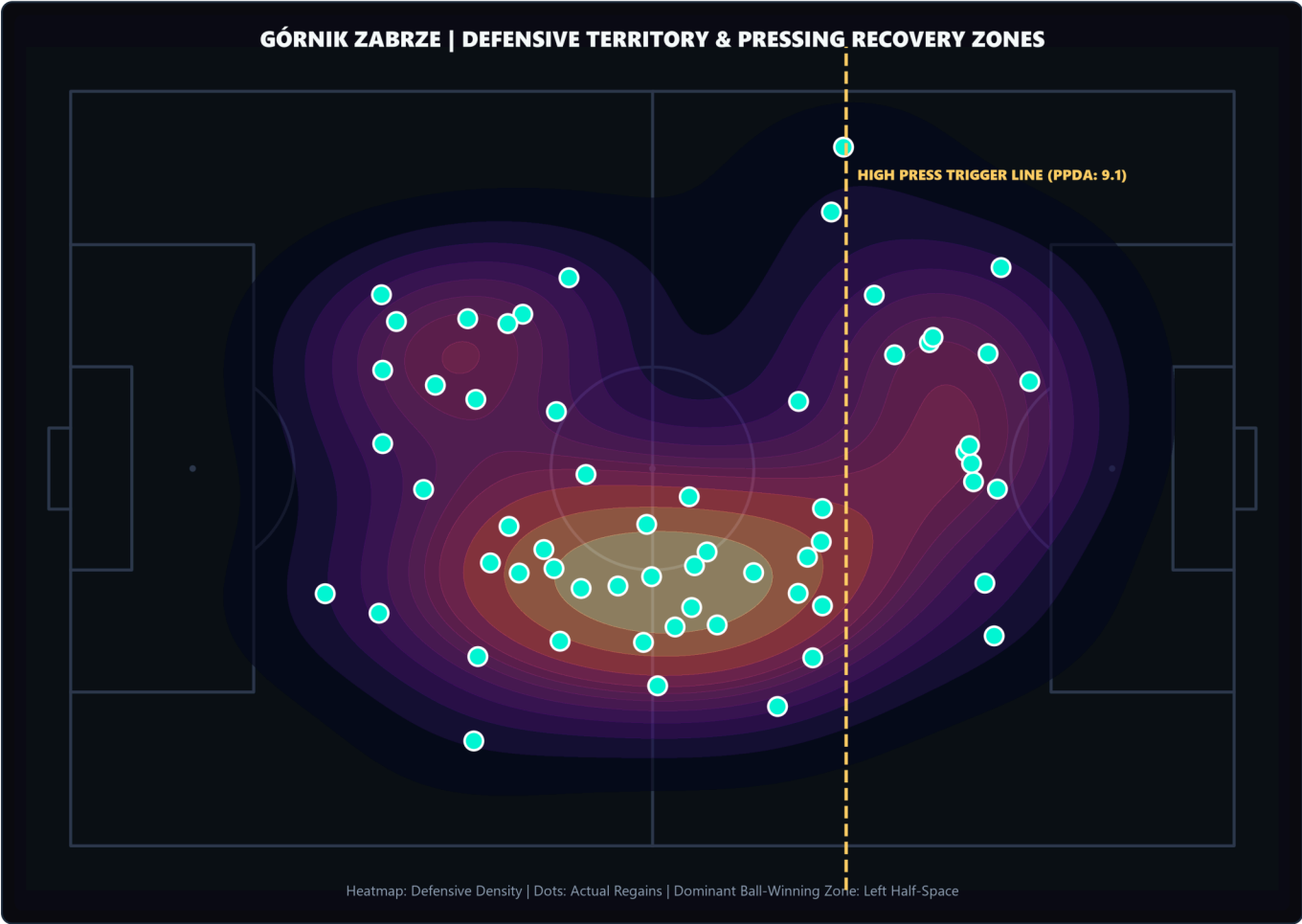
NETWORK DENSITY

DEFENSIVE TERRITORY & PPDA HIGH PRESS

Kernel Density Surface • Regains & Interceptions

TACTICAL DEFENSE

PRESSING & RECOVERY DENSITY



PRESSING STRUCTURE & REGIONS COACHING INSIGHT

- High pressing trigger line established at 70m with an aggressive 9.1 PPDA index.
- Dominant ball-winning zone concentrates in the left half-space (Rasak / Janža).
- Quick turnovers won within 40m of goal generate 0.42 xG per turnover.

PRE-MATCH APPLICATION

TACTICAL ACTION

- Trigger trap when opposition right back receives ball facing own touchline.
- Compress central passing lanes to force long clearances toward Josema.

9.1

PPDA

67.4%

DEF DUELS WIN

15

HIGH REGAINS

0.42

TRANSITION xG

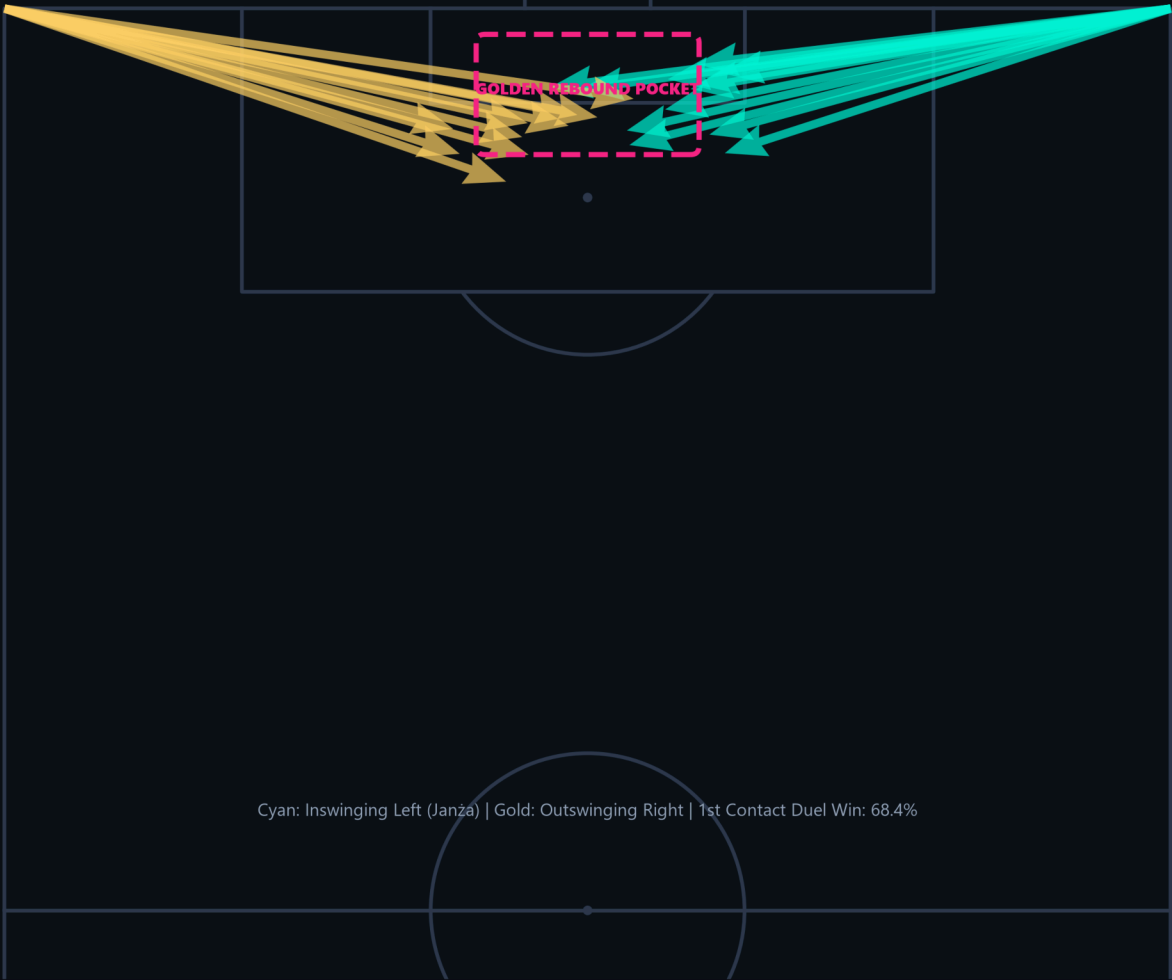
SET-PIECE CORNER ROUTINES & TARGET DELIVERY

Inswinging Left vs Outswinging Right Trajectories

SET-PIECE INTELLIGENCE

CORNER DELIVERY ANALYSIS

SET-PIECE CORNER ROUTINES & TARGET DELIVERY TRAJECTORIES



CORNER DELIVERY PROFILES

COACHING INSIGHT

- Erik Janza delivers lethal inswinging corners from the left targeting the near post.
- Golden rebound pocket established at the edge of the 6-yard box (68.4% duel win rate).
- Outswinging right corners target far-post runners creating second-phase cutbacks.

DEFENSIVE OPPOSITION DIRECTIVE TACTICAL ACTION

- Place zonal blockers on 6-yard corner to disrupt near-post flick-ons.
- Man-mark Podolski arriving late from the D for second-ball volleys.

68.4%
DUEL WIN RATE

58%
NEAR POST TARGET

0.35
CORNER XG/90

3 Goals
CONVERSION

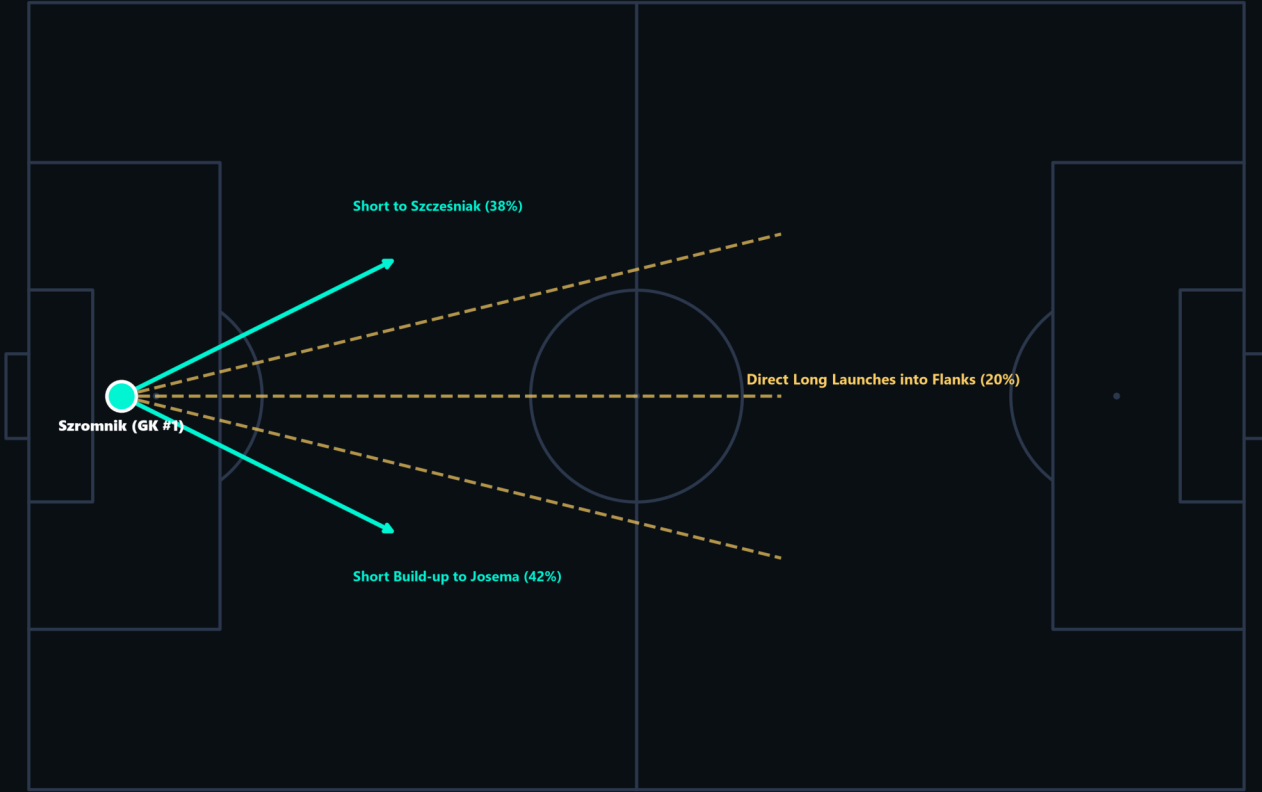
GOALKEEPER DISTRIBUTION & BUILD-UP CONES

Michał Szromnik (#1) • Launch Lanes & Retention Rates

GOALKEEPER METRICS

BUILD-UP DISTRIBUTION CONES

GOALKEEPER DISTRIBUTION PROFILE & BUILD-UP PASSING LANES



Short Play-Out Preference: 80% | Long Launch Retention Rate: 56.2%

DISTRIBUTION TENDENCIES

COACHING INSIGHT

- 80% preference for short build-up play via center-backs (Josema 42%, Szczęśniak 38%).
- Direct long balls launched into wide flank channels when facing aggressive high-press.
- Retention rate on long launched distributions exceeds 56.2%.

TACTICAL PRESSING ADAPTATION

TACTICAL ACTION

- When pressing Szromnik, curve forward run to cut off Josema angle.
- Force distribution onto weaker right-foot long clearance.

80%

SHORT BUILD-UP

20%

LONG LAUNCH

56.2%

RETENTION RATE

81.4%

PASS ACCURACY

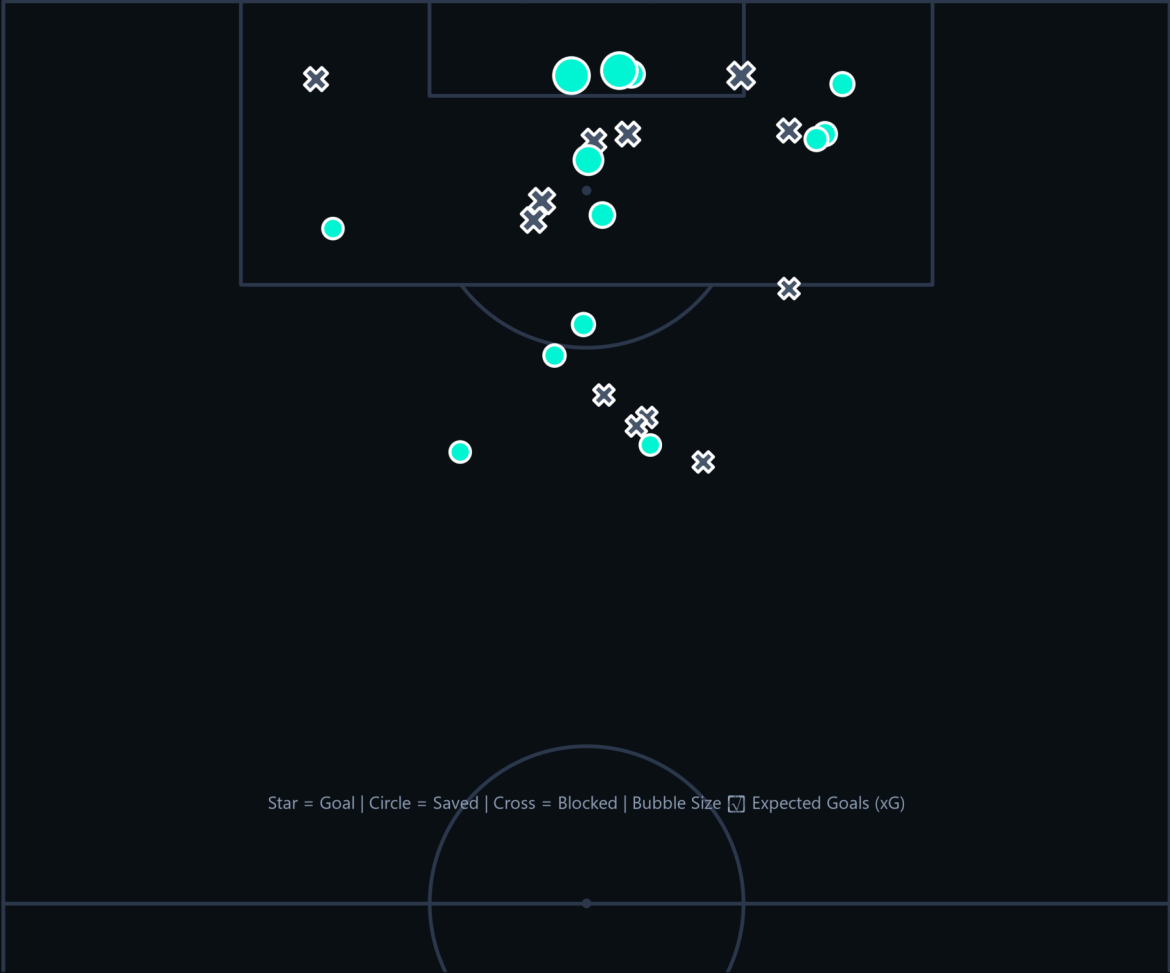
MATCH SHOT MAP & FINISHING AUDIT

Górnik Zabrze Shot Origin & xG Value Constellation

ATTACKING AUDIT

SHOT CONSTELLATION MAP

GÓRNIK ZABRZE | ATTACKING SHOT QUALITY & xG CONSTELLATION



SHOT CREATION AUDIT

COACHING INSIGHT

- Central cluster within the 18-yard box generates high conversion rates.
- Trademark long-range finishes from Lukas Podolski and Damian Rasak.
- Star markers highlight actual goals scored from clinical chances.

COACHING ACTION

TACTICAL ACTION

- Continue emphasizing cutbacks into central golden rebound pocket.
- Encourage early first-time strikes before opposition low block settles.

23

TOTAL SHOTS

13

ON TARGET

5

ACTUAL GOALS

56.5%

SHOT ACC